# The closure criterion for handedness — and why the biconditional it was framed as is false even over groups

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The decisive replacement for the vacuous Proposition 5.1 of the positive-home note (`2026-10-19-handedness-positive-home`), requested by Rick's referee report (M3/Q2, artifact `proofs/reviews/2026-10-21-triangulation-rick.md`). Registers in `sesquilaw-magmoid-karoubi.json` under node `positive-home-one-sided-absorbing-idempotent`, replacing child `principal-normalizer-always-associative`.

## 1 What this proves, and the correction it carries

Rick identified the structural "why" behind Theorem B of the positive-home note ($N_K$ associative $\iff$ the normalizer $K$ is two-sided absorbing): it is a **closure criterion**. This note states and proves it cleanly, in the (LA)-normalizer notation, as an explicitly attributed specialization of the classical transversal / right-loop construction.

Two results, one positive and one corrective:

- **Theorem 1 (closure criterion — the decisive statement).** If $\operatorname{Im} K$ is closed under composition $\cdot$, then $\# = \cdot|_{\operatorname{Im} K}$ and $N_K$ is associative. Contrapositive — *the decisive statement Rick asked for*: **if $N_K$ is handed (non-associative), then $\operatorname{Im} K$ is NOT closed under $\cdot$.** Airtight; it uses only idempotence of $K$ and associativity of $\mathcal{A}$.
- **Theorem 2 (the converse FAILS — sharpness, and a correction to the record).** The converse "$\operatorname{Im} K$ not closed $\Rightarrow$ handed", equivalently "$N_K$ associative $\Rightarrow$ $\operatorname{Im} K$ closed", is **false** — and it fails *even over invertible (group) bases*. Hence the biconditional stated as the target in `state/PROVE.md` ("handedness $\iff$ (LA)$\wedge$ non-closure") and the memory attribution "# associative $\iff$ $\operatorname{Im} K$ closed = classical right-loop group-iff-subgroup" are **incorrect**: that biconditional is not a theorem. Non-closure is **necessary but not sufficient** for handedness. The honest content is the one-directional Theorem 1.

The genuine biconditional sitting underneath is $\operatorname{Im} K$ **closed** $\iff$ $\# = \cdot|_{\operatorname{Im} K}$ (Prop 3) — closure is equivalent to the normalized product *coinciding* with the ambient product, not to associativity.

**Grades.** Theorem 1 and Prop 3 are **proved** (elementary, by hand, re-derived below and verified on an exact finite model). Theorem 2's refutation is **proved** by two explicit exact counterexamples (a minimal order-3 non-invertible monoid; and $C_4/D_4/Q_8$ normal-subgroup transversals), computed exactly in `fractions`/finite enumeration. Attribution is to the classical transversal literature (Baer, Lal, Kiechle, Grabowski), with the explicit note that the folklore *biconditional* those sources are sometimes quoted for does not hold.

## 2 Setting and the standing definition

The definitions are those of the positive-home note (§1), repeated for self-containment.

**Definition 2.1** (normalizer, normalized magmoid)**.** Let $(\mathcal{A}, \cdot, 1)$ be a category (composition diagrammatic: $f \cdot g$ = "$f$ then $g$"). A **normalizer** is a family $K = K_{X,Y} : \mathcal{A}(X,Y) \to \mathcal{A}(X,Y)$ with

- **(K1)** $K$ idempotent: $K(K(a)) = K(a)$;
- **(U)** $K(1_X) = 1_X$;
- **(LA)** left-absorption: $K(K(x) \cdot c) = K(x \cdot c)$ for all composable $x, c$.

The **normalized magmoid** $N_K$ has $\operatorname{Ob} N_K = \operatorname{Ob} \mathcal{A}$, $N_K(X,Y) := \operatorname{Im} K_{X,Y} = \{a : K(a) = a\}$, product $f \# g := K(f \cdot g)$, units $1_X$. $K$ is **two-sided** if $(\mathrm{RA}_\#)$: $K(f \cdot K(g \cdot h)) = K(f \cdot g \cdot h)$ for all $f, g, h \in \operatorname{Im} K$. $N_K$ is **handed** if it is non-associative (equivalently, under (LA), (LA) $\wedge \neg(\mathrm{RA}_\#)$ — Theorem B of the positive-home note).

By (K1), $\operatorname{Im} K = \{a : K(a) = a\}$ (if $a = K(x)$ then $K(a) = K(K(x)) = K(x) = a$; conversely immediate). We say $\operatorname{Im} K$ **is closed under** $\cdot$ if $f, g \in \operatorname{Im} K$ (composable) $\Rightarrow f \cdot g \in \operatorname{Im} K$, i.e. $K(f \cdot g) = f \cdot g$.

**Attribution (read before the theorem).** This whole package is the classical **transversal / right-loop** construction. For a subgroup $H \leq G$ with right transversal $T$ (one representative per right coset $Hg$, with $1 \in T$), the representative map $r : G \to T$, $r(g) =$ the $t \in T$ with $Hg = Ht$, is a retraction onto $T$ satisfying exactly (K1), (U), (LA); the induced product $t \circ t' := r(t \cdot t')$ is our $\#$, and $T = \operatorname{Im} r$. Sources: **Baer, "Nets and groups," Trans. AMS 46 (1939) 110–141**; **Lal, "Transversals in groups," J. Algebra 181 (1996) 70–81**; **Kiechle, *Theory of K-Loops*, Springer LNM 1778 (2002)**; **arXiv:1309.7729, "On the Congruences in Right Loops"**; oidified: **Grabowski, "An introduction to loopoids," arXiv:1510.03395** (Def 4.1 / Prop 4.3: $t \bullet t' = \pi(t \cdot t')$). The forward direction of Theorem 1 ("$T$ a subgroup $\Rightarrow \circ = \cdot|_T$") is classical and elementary. The *converse* — the folklore biconditional "right loop is a group $\iff$ transversal is a subgroup" — is **not** a theorem (§3).

## 3 Theorem 1 — the closure criterion

**Theorem 1** (closure criterion)**.** *Let $K$ be a normalizer on $\mathcal{A}$. If $\operatorname{Im} K$ is closed under $\cdot$, then:*

*(1) $f \# g = f \cdot g$ for all composable $f, g \in \operatorname{Im} K$ (i.e. $\# = \cdot|_{\operatorname{Im} K}$);*

*(2) $N_K$ is associative.*

***Contrapositive (the decisive statement).*** *If $N_K$ is handed, then* $\operatorname{Im} K$ *is* ***not*** *closed under $\cdot$.*

*Proof.* (1) Let $f, g \in \operatorname{Im} K$ be composable. Closure gives $f \cdot g \in \operatorname{Im} K$, so $K(f \cdot g) = f \cdot g$. Hence $f\#g = K(f \cdot g) = f \cdot g$.

(2) Let $f, g, h \in \operatorname{Im} K$ be composable. By (1), $f\#g = f \cdot g$; closure gives $f \cdot g \in \operatorname{Im} K$, so with $h \in \operatorname{Im} K$, (1) applies again: $(f\#g)\#h = (f \cdot g)\#h = (f \cdot g) \cdot h$. Symmetrically, $g\#h = g \cdot h \in \operatorname{Im} K$, so $f\#(g\#h) = f\#(g \cdot h) = f \cdot (g \cdot h)$. By associativity of $\cdot$ in $\mathcal{A}$, $(f \cdot g) \cdot h = f \cdot (g \cdot h)$. Therefore $(f\#g)\#h = f\#(g\#h)$.

The contrapositive is immediate. □

*Remark* 3.1 (the proof uses almost nothing). Theorem 1 invokes only **(K1)** (to read $\operatorname{Im} K$ as the fixed-point set) and **associativity of $\cdot$ in $\mathcal{A}$**. It does *not* use (U) or even (LA). So closure is a sufficient condition for associativity that is insensitive to the left-absorption law — a first sign that it is strictly stronger than what associativity needs (§3). In the standing setting (LA) holds, so "handed" means (LA) $\wedge \neg(\mathrm{RA}_\#)$; Theorem 1 gives (LA) $\wedge \neg(\mathrm{RA}_\#) \Rightarrow$ not closed. This is exactly Rick's "handed $\Rightarrow \operatorname{Im} K$ not closed."

*Remark* 3.2 (relation to Theorem B). Theorem 1 is the structural "why" behind Theorem B of the positive-home note. Indeed closure $\Rightarrow N_K$ associative $\iff (\mathrm{RA}_\#)$ (Theorem B). So **closure** $\Rightarrow (\mathrm{RA}_\#)$. The content of §3 is that this implication is *strict*: $(\mathrm{RA}_\#)$ can hold without closure.

# 4 Theorem 2 — the converse fails, even for groups

The target in `state/PROVE.md` asked for the biconditional "handedness $\iff$ non-closure", equivalently "$N_K$ associative $\iff$ $\operatorname{Im} K$ closed". Only the $\Leftarrow$ of the second form (= Theorem 1) is true. The $\Rightarrow$ is false.

**Theorem 2** (sharpness)**.** *There exist normalizers $K$ (satisfying (K1), (U), (LA)) with* $\operatorname{Im} K$ ***not*** *closed under $\cdot$ and yet $N_K$* ***associative****. Such examples exist already (a) on a 3-element monoid — the minimal order — with a non-invertible element, and (b) on finite* ***groups*** *$C_4, D_4, Q_8$. Hence non-closure is* ***necessary but not sufficient*** *for handedness, and invertibility of the base does* ***not*** *restore the converse.*

## 4.1 A minimal non-invertible witness (order 3)

Let $M = \{e, a, b\}$ be the monoid generated by one element $a$ with $a^3 = a$ (so $b := a^2$). Its table (rows = left factor):

| $\cdot$ | $e$ | $a$ | $b$ |
|---|---|---|---|
| $e$ | $e$ | $a$ | $b$ |
| $a$ | $a$ | $b$ | $a$ |
| $b$ | $b$ | $a$ | $b$ |

$M$ is associative with unit $e$ (both verified exactly). The element $a$ is **not invertible** ($a \cdot x = e$ has no solution; likewise $x \cdot a = e$), so $M$ is **not a group**.

Define the normalizer $K$ by $K(e) = e$, $K(a) = a$, $K(b) = a$. Then (checked over all $3 \times 3$ pairs, exactly):

- (K1) $K \circ K = K$, (U) $K(e) = e$, and (LA) $K(K(x) \cdot c) = K(x \cdot c)$ hold **universally**.
- $\operatorname{Im} K = \{e, a\}$, and it is **not closed**: $a \cdot a = b \notin \operatorname{Im} K$.

- $N_K$ on $\{e, a\}$ has table $e\#e = e$, $e\#a = a\#e = a$, $a\#a = K(b) = a$ — the 2-element semilattice, which is **associative** (all 8 triples).

So (LA) holds, $\operatorname{Im} K$ is not closed, yet $N_K$ is associative: the converse of Theorem 1 is refuted. An exhaustive search over all monoids of order $\leq 3$ and all normalizers on them shows **no such example exists at order** $\leq 2$ (0 pairs), and exactly **6** exist at order 3 (across 3 monoid tables) — so order 3 is minimal.

**What goes wrong for the converse here.** $\operatorname{Im} K = \{e, a\}$ fails closure only at the single product $a \cdot a = b$, whose normalization $K(b) = a$ "lands back" compatibly: the one escaping product does not propagate to any 3-fold associativity defect. Non-closure is a *local* (binary) defect; handedness is a *global* (ternary) defect, and the former does not force the latter.

### 4.2 Group witnesses ($C_4$, $D_4$, $Q_8$) — invertibility does not help

Take $K = r$, the right-coset representative map of a subgroup $H \leq G$ with transversal $T$ (so $\operatorname{Im} K = T$, $\# = \circ$). (LA) holds in every case. The biconditional "$\circ$ associative $\iff$ $T$ subgroup" **fails**:

- $C_4 = \mathbb{Z}/4$**,** $H = \{0, 2\}$**,** $T = \{0, 1\}$**.** Representatives: $r(0) = r(2) = 0$, $r(1) = r(3) = 1$. $\operatorname{Im} K = \{0, 1\}$ is **not closed** ($1 + 1 = 2 \notin T$). But $(T, \circ)$: $1 \circ 1 = r(2) = 0$, so $(T, \circ) \cong C_2$ is a **group** — associative.
- $D_4$ (order 8): 12 such transversals; $Q_8$: 20, e.g. $H = \{\pm 1\}$, $T = \{1, -i, -j, -k\}$ with $(-i)(-j) = k \notin T$, yet $(T, \circ) \cong C_2 \times C_2$ associative.

**Mechanism (clean and general).** Whenever $H \triangleleft G$ is normal, the quotient map restricts to a bijection $T \to G/H$, $t \mapsto Ht$, under which $\circ$ becomes the group law of $G/H$: $r(t \cdot t') \mapsto H(t \cdot t') = (Ht)(Ht')$. So $(T, \circ) \cong G/H$ is a group for **every** transversal of a normal subgroup — while a transversal of a nontrivial normal subgroup is almost never a subgroup. This is a whole family of counterexamples; the vanishing of anomalies for $C_2, C_3, C_2 \times C_2, S_3$ in the sweep is an accident of those small subgroup lattices (every transversal of each normal subgroup there happens to be a subgroup), not a theorem.

This is the decisive correction: **invertibility does not rescue the converse.** The folklore "right loop is a group $\iff$ transversal is a subgroup" is simply false in the presence of a nontrivial normal subgroup. The classical literature's *true* content is the forward direction ("subgroup $\Rightarrow \circ = \cdot|_T$") — i.e. Theorem 1 — not the biconditional.

## 5 Proposition 3 — the biconditional that *is* true

**Proposition 1.** *For a normalizer $K$: $\operatorname{Im} K$ is closed under $\cdot$* ***if and only if*** *$\# = \cdot|_{\operatorname{Im} K}$ (i.e. $f\#g = f \cdot g$ for all composable $f, g \in \operatorname{Im} K$).*

*Proof.* ($\Rightarrow$) Theorem 1(1). ($\Leftarrow$) If $f\#g = f \cdot g$ for all $f, g \in \operatorname{Im} K$, then $f \cdot g = f\#g = K(f \cdot g) \in \operatorname{Im} K$, so $\operatorname{Im} K$ is closed. □

So the honest "iff" pairs closure with **coincidence of the two products**, not with associativity. Associativity is a strictly weaker ternary condition:

$$\begin{aligned} \operatorname{Im} K \text{ closed} &\iff \# = \cdot|_{\operatorname{Im} K} \qquad \text{(Prop 3)} \\ &\Downarrow \text{ (strict, by Thm 2)} \\ (\mathrm{RA}_{\#}) \text{ holds} &\iff N_K \text{ associative} \qquad \text{(Theorem B)} \end{aligned}$$

The classical transversal statement, stated correctly, is the top line specialized to groups: "$T$ is a subgroup $\iff$ $\circ$ is the restriction of $\cdot$." The middle-arrow strictness is the whole content of §3.

*Remark* 5.1 (the obstruction, and the resonance with the weld $H^2$ program). For $x, y \in \operatorname{Im} K$ write the de-normalization $x \cdot y = \varphi(x, y) \cdot (x \# y)$ where $\varphi(x, y)$ is whatever $K$ strips off ($K(x \cdot y) = x \# y$). Then closure $\iff$ $\varphi \equiv 1$ (trivial factor). Associativity of $\#$ is a **2-cocycle-type equation** on $\varphi$, which $\varphi$ can satisfy *without* being trivial — precisely the normal-subgroup case, where $\varphi$ is (the data of) a split extension and $(T, \circ) \cong G/H$. This is why closure $\Rightarrow$ associativity but not conversely, and it is the same split-vs-nonsplit dichotomy that the weld obstruction lives in (`h2-weld-class-is-splitting-not-associativity`): closure is the *trivial-factor* condition, strictly below the *cocycle* condition. The handedness home Norm of (`2026-10-19-handedness-positive-home`) realizes the strict gap from the other side — there $\operatorname{Im} K$ is not closed **and** $(\mathrm{RA}_{\#})$ fails.

## 6 The corrected minimality statement (replacing Prop 5.1)

Rick showed the old Proposition 5.1 ("a principal normalizer $K(x) = \kappa \cdot x$ by an idempotent *element* is always associative") is vacuous under (U): $K(1) = \kappa$ forces $\kappa = 1$, $K = \mathrm{id}$. The **decisive replacement** is:

**Corollary 1** (home of handedness)**.** *If $N_K$ is handed then* $\operatorname{Im} K$ *is not closed under $\cdot$ (Theorem 1). Consequently the home of handedness is a normalizer whose image is* ***not*** *a subcategory of $\mathcal{A}$. Non-closure is* necessary*; by Theorem 2 it is* not sufficient*, so the full characterization remains* $(\mathrm{LA}) \wedge \neg(\mathrm{RA}_{\#})$ *(Theorem B), of which non-closure is the clean necessary shadow.*

The Norm instance realizes it: $\operatorname{Im} K$ = row-stochastic kernels is **not closed** under matrix product in the sense required — the product of two normalized kernels de-normalizes (mass is destroyed at $\perp$ states before re-normalization), so $K(f \cdot g) \neq f \cdot g$ in general, and additionally $(\mathrm{RA}_{\#})$ fails (exact witness, re-verified 2026-10-21: $(f\#g)\#h = [8/11, 3/11] \neq [2/3, 1/3] = f\#(g\#h)$). Non-closure is the necessary half; the sufficient defect is the ternary $(\mathrm{RA}_{\#})$ failure.

## 7 Verification (computational, exact)

Scripts under `scratch/`, all exact (`fractions.Fraction` / finite enumeration), 2026-10-21:

- `2026-10-21-task1-order3-converse-counterexample.py`: the order-3 witness — (LA) universal, $\operatorname{Im} K$ not closed ($a \cdot a = b$), $N_K$ associative. Converse refuted.
- `2026-10-21-task2-minimal-order-search.py`: exhaustive order $\leq 3$ — 0 witnesses at order $\leq 2$, 6 at order 3. Minimality confirmed.
- `2026-10-21-task3-group-converse-transversals.py`: $C_4/D_4/Q_8$ normal-subgroup transversals with (LA) holding, $\circ$ associative, $T$ not a subgroup. Biconditional refuted over groups. (Mechanism $(T, \circ) \cong G/H$ confirmed.)
- `2026-10-21-task4-norm-witness.py`: Norm handedness witness re-verified exactly.

## 8 Gaps / honest scope

- **Theorem 1 and Prop 3 are the proved core** (elementary, by hand; (LA) not even required for Theorem 1). These are the registrable results.
- **Theorem 2** is a refutation by explicit exact examples; the general "normal $H \Rightarrow (T, \circ) \cong G/H$" mechanism is classical (quotient group), stated as the reason, not a new theorem.
- **The record correction** (the folklore "group $\iff$ subgroup" biconditional is false; the memory note and `PROVE.md` target over-stated it) is established by §3 and should propagate to the registry and to `feedback-handedness-is-classical-transversal-not-original`.
- No claim of originality beyond: (i) the base is a **general associative category/monoid without invertibility** (the order-3 witness is non-invertible); (ii) the clean **statement of the strict gap** closure $\subsetneq (\mathrm{RA}_{\#})$ and its refutation of the folklore biconditional *even over groups*; (iii) the Norm substochastic-kernel instance. The closure criterion itself (forward direction) is classical, cited above.